

Sudarshan R. Hegde

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CAREER OBJECTIVE

Highly motivated and detail-oriented Computer Science Engineering student with strong foundations in data structures and algorithms. Proficient in C++, Java, Python, and modern web technologies, with a passion for applying machine learning and AI to solve real-world problems, with a focus on continuous learning and impactful contributions.

TECHNICAL SKILLS

- **Languages:** C++, C, Java, Python, JavaScript
- **Frontend:** HTML, CSS, React, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Cloud/DevOps:** AWS, Azure, Google Cloud, Jenkins
- **Data Science/ML:** TensorFlow, PyTorch
- **Databases:** MySQL
- **Tools:** Git, VS Code, PyCharm, LaTeX
- **Systems:** Ubuntu, Linux, Windows, Maven, Gradle

WORK EXPERIENCE

Product Development Intern, Continuum Associates LLC (Remote / India)

Jan 2026 – Present

Contributing to the development of internal technology products and decision-support tools for clients in the electric power and energy sector, focusing on data-driven analysis and intuitive user interfaces. Collaborating with cross-functional teams to translate quantitative analysis and planning studies into robust software features, ensuring alignment with ISO, RTO, and industry planning rules. Assisting in prototyping, testing, and iterating on product features that support high-impact engineering and strategy studies, emphasizing performance, reliability, and user-centric design.

Artificial Intelligence with Cloud Computing Internship, SuprMentr Technologies Pvt Ltd

Jan 2026 - May 2026

Focused on developing and deploying AI models within cloud-native environments, I leverage scalable infrastructure to optimize performance and cost-efficiency. I collaborate on architecting automated data pipelines and integrating machine learning services into cloud platforms, ensuring the delivery of robust, high-availability solutions for complex data challenges.

Software Developer Intern, Heartiest Mind Technologies Pvt Ltd

Jan 2026 - April 2026

Focused on the full software development lifecycle, I contribute to the design, testing, and deployment of scalable applications. I collaborate with cross-functional teams in an Agile environment to build robust features, optimize system performance, and implement clean, efficient code to meet client-specific requirements.

Creative Team Member - OSGeometry-SaIT Chapter, OSGeometry

Sept 2025 - May 2026

Operating at the intersection of design and development, I lead OSGeometry's visual strategy while contributing directly to open-source repositories. My role encompasses UI/UX design, codebase maintenance, and technical documentation, ensuring a seamless and intuitive experience for the developer community.

PROJECTS

Geo-Agri-Analyst

PyTorch, React.JS, Python, Kaggle, FastAPI, Hugging Face

Geo-Agri Analyst is an end-to-end deep learning pipeline that enhances low-resolution satellite imagery using custom Super-Resolution GANs to train high-accuracy land-use classifiers with minimal labeling via Active Learning strategies. It features a deployable web application that fetches live Sentinel-2 data to provide real-time agricultural insights and land classification.

Portfolio Website

React.js, Tailwind CSS, GitHub Pages

Designed and deployed a personal portfolio featuring responsive layouts and reusable components. Showcases projects, certifications, and contact details.

Polygon Color Filling using UNet

PyTorch, Kaggle

Developed a UNet-based deep learning model to fill polygon shapes with specified colors from text input. Integrated custom dataset loading, model training, and evaluation with Weights & Biases tracking.

CERTIFICATIONS

- Deep Learning – IIT Ropar [NPTEL]
- Robotic Process Automation (RPA)
- Power BI and Data Visualization Skill Development Program
- Data Analytics with Power BI Workshop

EDUCATION

B.E. in Computer Science and Engineering

2022 – 2026

Sambhram Institute of Technology, VTU

CGPA: 9.1 / 10

PCMCS

2020 – 2022

MES PU College, Sirsi, Karnataka

Grade: 88%

SSLC

2018 – 2020

Shri Kalika Bhavani EM High School, Kansur, Karnataka

Grade: 94%

PUBLICATIONS

Geo-Agri Analyst: An End-to-End Decision Support System for Agriculture via Super-Resolution and Active Learning

Published in International Journal of Scientific Development and Research (IJS DR), Vol. 11, Issue 2 Feb 2026

Developed a unified deep learning framework to democratize precision agriculture by synergizing a **Perceptual Extreme Super-Resolution Network (RFB-ESRGAN)** with a **Deep Bayesian Active Learning** strategy. The system achieves a fourfold increase in the spatial resolution of Sentinel-2 multi-spectral imagery, effectively recovering latent high-frequency textures for granular land cover classification. By implementing a hybrid query strategy combining Distance-Based Sampling and Spatial-Spectral Entropy, the pipeline reduces labeled data requirements by approximately **85%** while improving Top-1 classification accuracy by **5.4%**. Validated on accessible cloud infrastructure using the BigEarthNet archive, the project was deployed as a full-stack solution using **PyTorch, FastAPI, and React**. Read Full paper at <https://ijsdr.org/papers/IJS DR2602037.pdf>

LANGUAGES

English, Kannada and Hindi